

Knee Joint Safety Demonstrator

System Safety Engineering Demonstrator

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1. System Definition

1.1 Purpose

This project looks at one load-bearing knee joint of an industrial humanoid robot.

The joint has to convert a motion or torque request from the robot controller into controlled knee movement, while still being able to detect local faults and react before they develop into hazardous motion.

The aim is not to design the complete humanoid robot, but to understand how requirements, architecture, diagnostics and fault reactions can be built around one safety-critical actuator.

1.2 System Boundary

The system under consideration is the complete local knee-joint actuation chain.

Included inside the boundary:

- Communication interface
- Local joint controller
- Motor-control function
- Gate driver
- Three-phase inverter
- PMSM motor
- Reduction gearbox / transmission
- Motor encoder
- Joint-output encoder
- Motor-current sensing
- Voltage and temperature monitoring
- Local safety monitoring
- Independent torque-disable path
- Optional joint brake

The physical joint output is included because the safety concern is not only what the electronics command, but what the knee actually does.

1.3 External Systems

The knee system interacts with the following robot-level systems.

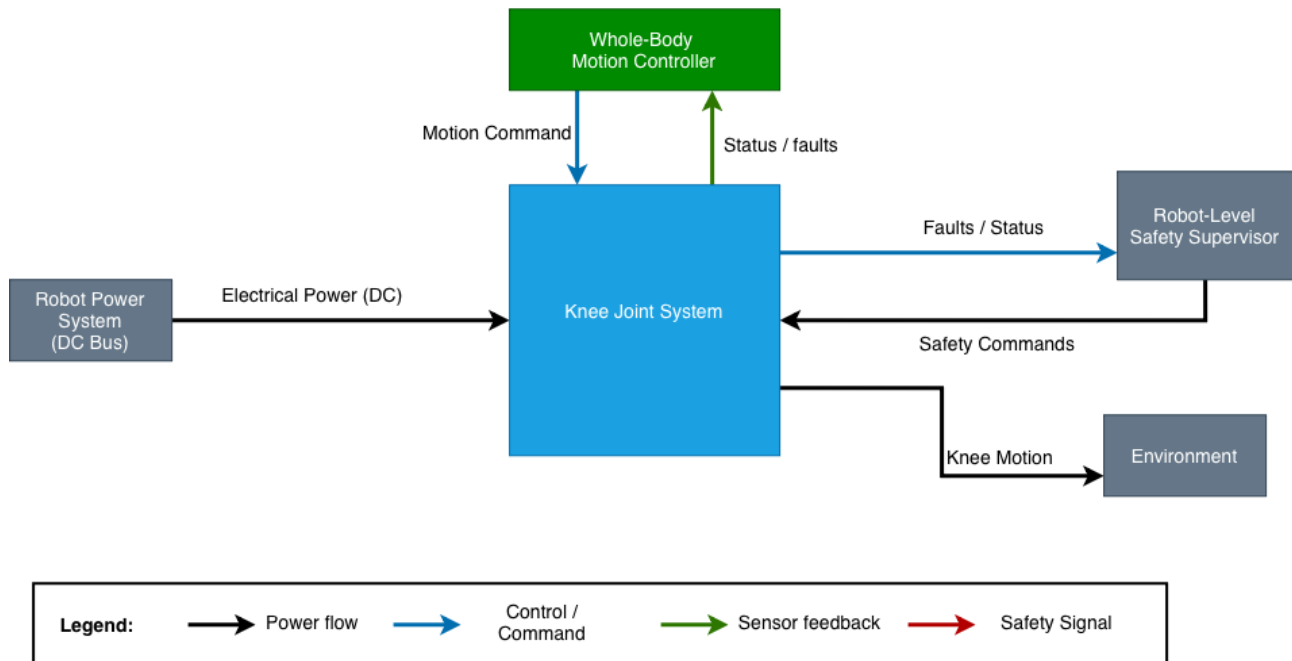
Whole-Body Motion Controller. Provides the requested joint behavior, for example desired joint position, velocity, torque, operating mode and enable / disable request. It is outside the knee system boundary because it coordinates the complete robot rather than only the knee.

Robot Safety Supervisor. Coordinates robot-level reactions such as controlled stop, degraded operation, posture recovery and higher-level power isolation. The knee system reports fault and health status to this supervisor.

Robot Power System. Provides the actuator DC supply. For this demonstrator, a nominal actuator DC bus of approximately 48 V is assumed.

Robot Mechanical Structure and Environment. The knee is connected to the thigh and lower leg and is influenced by robot weight, payload, inertia, ground-reaction forces, disturbances from other joints and contact with the environment. These loads are treated as external inputs to the knee system.

Figure 1 - System Context Diagram



1.4 Main Inputs

The knee system receives:

- Desired joint position
- Desired joint velocity
- Desired joint torque
- Mode / enable command
- DC electrical power
- Mechanical load at the joint
- Robot-level safety requests

Local sensor feedback is treated as an internal input to the control and safety functions.

1.5 Main Outputs

The system produces:

- Knee joint torque
- Knee joint angular motion

- Actual joint position and velocity
- Actuator status
- Diagnostic information
- Detected fault status
- Degraded / safe-state status

1.6 Functional Chain

The nominal control path is:

Whole-Body Controller -> Communication Interface -> Local Joint Controller -> Motor Control -> Gate Driver -> 3-Phase Inverter -> PMSM Motor -> Gearbox -> Knee Joint Output

The corresponding energy path is:

Robot Power System -> DC Bus -> Inverter -> Motor -> Gearbox -> Knee Joint

The feedback path includes the motor encoder, joint encoder, current sensing, voltage sensing and temperature sensing. These signals are used by both the normal control path and the safety-monitoring path.

1.7 Operating Scenario Used in This Project

The reference scenario is a humanoid walking slowly near a person while carrying a load. The knee is load-bearing and contributes to maintaining posture and controlled forward motion.

A fault causes the knee to move faster than intended or produce unintended / excessive torque. Possible consequences include:

- Loss of controlled walking
- Collision with the nearby person
- Loss of balance or robot fall
- Dropped or destabilised payload
- Damage to the robot or nearby equipment

1.8 Important Safety Assumption

Immediate torque removal is not assumed to be safe by default.

Torque-off -> loss of joint support -> possible collapse

1.9 Initial Design Assumptions

- The actuator uses a PMSM motor and a three-phase inverter.
- A mechanical reduction stage is present between the motor and the knee output.
- Motor position and joint-output position are measured separately.
- Motor current is measured and available for control and diagnostics.
- The local knee controller cannot assume that commands from the whole-body controller are always correct.
- The safety-monitoring function should not rely completely on the same control path that may create the hazardous behavior.
- A single sensor fault should not automatically result in unrestricted actuator operation.

- Mechanical faults remain part of the safety analysis even if they cannot be fully mitigated electronically.

1.10 Out of Scope

The following are not developed in detail:

- Complete humanoid balance control
- Gait planning
- Perception / AI safety
- Battery-pack safety
- Full robot structural analysis
- Cybersecurity
- Complete brake design
- Full certification assessment
- Detailed motor electromagnetic design

These may appear as external interfaces or assumptions, but they are not the focus of this demonstrator.

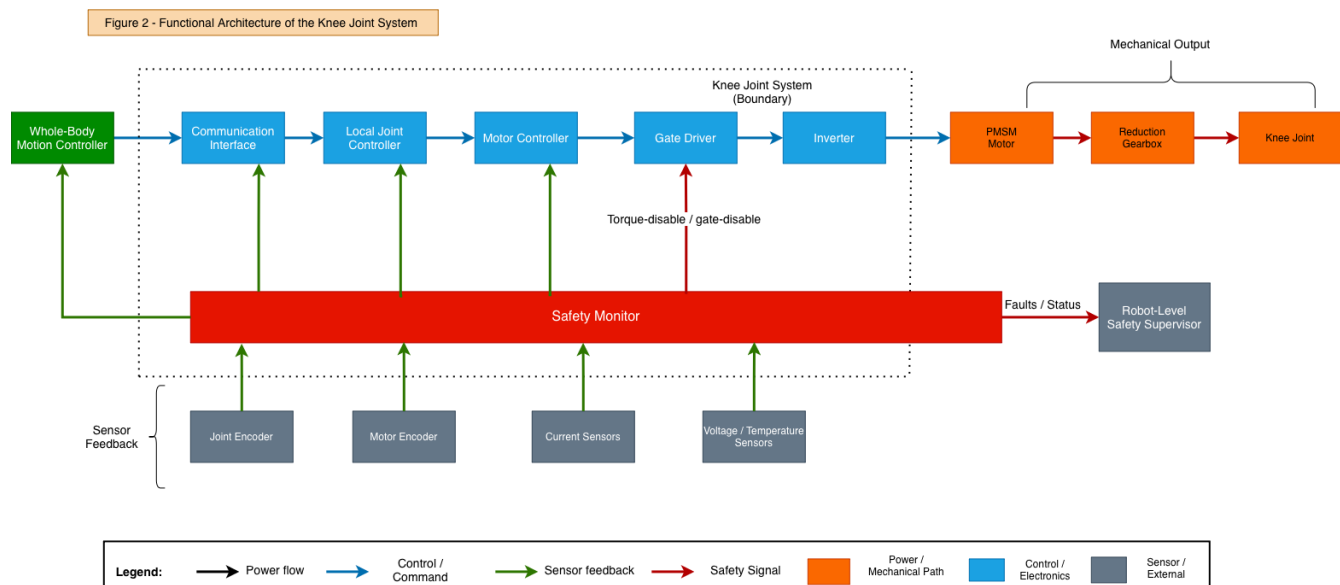
2. System Architecture

2.1 Architecture Objective

The knee-joint architecture was developed around one main safety concern: unintended or excessive knee torque or motion while the joint is load-bearing.

The architecture therefore needs to support both normal closed-loop actuator control and local detection of abnormal behaviour. A key consideration is that the same control path that produces joint torque should **not** be the only path available to detect and mitigate hazardous torque.

The architecture is intentionally limited to the level required for this demonstrator. Detailed circuit design, processor selection, motor electromagnetic design and production-level redundancy concepts are outside the current scope.



2.2 Architecture Overview

The joint is split conceptually into four main paths:

- command and control
- electrical and mechanical energy flow
- sensor feedback and diagnostics
- safety monitoring and fault reaction

The command path determines what the actuator is requested to do.

The energy path is the physical path through which electrical energy becomes joint torque and motion.

The feedback path provides information about what the actuator is actually doing.

The safety path checks whether commanded and actual behaviour remain consistent and provides a means to react when they do not.

2.3 Command and Control Path

The normal command path is:

Whole-Body Motion Controller → Communication Interface → Local Joint Controller → Motor Controller → Gate Driver → Inverter

The Whole-Body Motion Controller provides joint-level motion intent.

The Local Joint Controller executes this request subject to local limits and diagnostics.

Motor Control / FOC regulates motor current and generates PWM for the inverter.

Allocation decision: whole-body motion planning remains at robot level, while fast actuator execution and monitoring remain local to the knee.

2.4 Energy and Mechanical Actuation Path

The physical energy path is:

Robot Power System → DC Bus → Three-Phase Inverter → PMSM Motor → Reduction Gearbox → Knee Joint

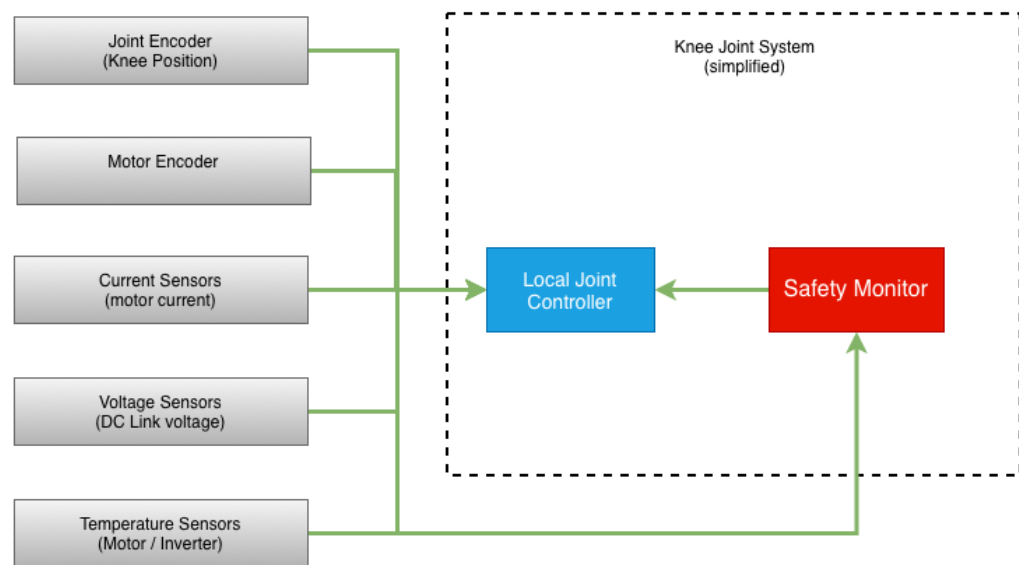
This is the physical hazard-propagation path. Faults in the power stage or transmission can cause joint behaviour to differ from the software request, so safety monitoring cannot rely only on command signals.

2.5 Feedback and Diagnostic Path

The baseline architecture uses feedback from several points in the actuator.

Signal	Used by	Architectural purpose
Joint encoder	Local Controller, Safety Monitor	Actual knee motion
Motor encoder	Motor Control, Safety Monitor	Rotor state + motor-side motion
Motor current	Motor Control, Safety Monitor	Electrical actuation / torque indication
Voltage / temperature	Controller, Safety Monitor	Operating-condition supervision

Figure 3 - Feedback and Diagnostic Signals



2.6 Cross-Monitoring Concept

The architecture uses measurements from different physical points to check whether the actuator behaves consistently.

One example is the relationship between motor-side and joint-side motion:

Motor Encoder ↔ expected gearbox relationship ↔ Joint Encoder

If the motor moves but the knee output does not follow the expected transmission relationship, the actuator state is inconsistent.

Possible causes include:

- gearbox or coupling fault
- motor encoder fault
- joint encoder fault
- incorrect scaling or configuration

The purpose of this check is not necessarily to identify the exact failed component immediately. The first objective is to detect that the measured actuator behaviour is no longer physically plausible.

A similar check is used in the electrical path:

Commanded Motor Current ↔ Measured Motor Current

A persistent difference indicates that the motor-current path is not following the requested actuation.

2.7 Safety Monitor

The architecture includes a Safety Monitor that observes selected safety-relevant relationships independently of the primary motion-control objective.

Relevant inputs include:

- commanded and measured joint velocity
- commanded and measured motor current
- motor encoder
- joint encoder
- communication status
- voltage and temperature status

The main monitoring functions are:

- joint-velocity deviation monitoring
- motor-current deviation monitoring
- motor-to-joint transmission plausibility
- communication supervision
- sensor plausibility
- controller/timing supervision

The Safety Monitor is not intended to be a duplicate joint controller.

Its purpose is to reduce the risk that a single fault in the normal control path can both create hazardous behaviour and prevent that behaviour from being detected.

The detailed implementation and required independence of this monitor are left open at this stage.

2.8 Independent Fault-Reaction Path

The architecture includes an independent torque-disable or gate-disable path.

This is needed because the normal controller itself may be the source of the fault.

For example:

Local Joint Controller Fault → incorrect current request → hazardous motor torque

If the same faulty controller is also the only means of stopping the motor, the architecture contains a single dependency.

The safety concept therefore assumes a separate hardware-capable path that can influence torque generation without relying completely on the normal control software.

The detailed implementation of this path is not fixed in this demonstrator.

2.9 Robot-Level Safety Supervisor

Local actuator faults are reported to a robot-level Safety Supervisor.

The local knee system is responsible for detecting actuator-level problems, while the robot-level supervisor coordinates actions involving the complete humanoid, such as:

- posture recovery
- coordinated stopping
- degraded whole-body operation
- higher-level power isolation

This split is intentional. The knee controller has the fastest access to local actuator information, but it does not have enough information to determine the safest response for the complete robot.

2.10 Fault-Reaction Philosophy

The architecture does not assume that every fault should result in immediate torque removal.

For a load-bearing knee, removing torque abruptly may cause loss of support and robot collapse.

The fault reaction therefore depends on which functions remain trustworthy after the fault.

If normal control is still considered reliable, the preferred response may be:

- torque limitation
- velocity limitation
- degraded operation
- controlled stop

If normal control can no longer be trusted, the reaction may require:

- independent torque disable
- power isolation

- mechanical brake
- coordinated robot-level recovery

The guiding principle is: Use the least disruptive reaction that still provides a trustworthy means of controlling the hazard.

2.11 Key Architecture Decisions

The following decisions were made during development of the demonstrator.

- Whole-body control remains outside the knee system boundary.
The whole-body controller coordinates the complete robot, while the knee system is responsible for local execution and monitoring.
- Motor-side and joint-side position are measured separately.
This provides information on both sides of the gearbox and supports transmission plausibility monitoring.
- Motor current is treated as safety-relevant feedback.
It provides information about the electrical torque-generation path and allows commanded-versus-actual actuation to be monitored.
- Safety monitoring is separated conceptually from normal control.
The objective is to avoid a single control-path failure defeating both control and fault mitigation.
- Immediate torque-off is not treated as the universal safe state.
The knee may be supporting robot weight, so fault reaction depends on remaining control capability.
- Mechanical failures remain inside the safety analysis.
An electrical safety concept alone cannot guarantee correct joint behaviour if the gearbox or coupling fails.

2.12 Assumptions and Open Points

The following points are assumptions for the current demonstrator and would require further engineering work in a real design:

- PMSM and three-phase inverter are assumed as the actuator technology.
- Approximately 48 V is used as a representative actuator DC bus, not as a derived requirement.
- The exact implementation and independence of the Safety Monitor are not yet defined.
- The exact design of the independent torque-disable path is not specified.
- Mechanical brake implementation remains optional.
- Exact diagnostic thresholds and reaction times are not yet derived.
- Sensor accuracy, diagnostic coverage and quantitative reliability targets Detailed common-cause and dependent-failure analysis remains future work.

These open points are kept explicit so that the architecture is not presented as more mature than it currently is.

3. Hazard Analysis

3.1 Purpose

The objective of this analysis is to identify the hazardous system behaviour that can result from unintended knee-joint actuation.

The selected safety concern is:

Unintended or excessive knee torque / motion while the joint is load-bearing.

3.2 Operating Scenario

The reference operating scenario is:

The humanoid robot is walking slowly near a person while carrying a load.

During this operation:

- the knee is load-bearing
- the robot is moving relative to the person
- the carried load adds additional mechanical energy and consequence
- knee torque is required for both motion and posture control

3.3 Malfunctioning Behaviour

The malfunctioning behaviour considered is:

The knee produces unintended or excessive torque or moves at a velocity inconsistent with the commanded joint behaviour.

3.4 Hazardous Event — HE-01

HE-01 — Unintended or excessive knee motion occurs while the humanoid is load-bearing and operating close to a person.

The unexpected knee behaviour may cause loss of controlled walking or posture.

Possible consequences include:

- collision with the nearby person
- loss of balance
- robot fall
- destabilisation or dropping of the carried load
- impact of the robot or payload with surrounding equipment

The severity of the outcome depends on robot mass, payload, joint velocity and the operating environment. These parameters are not quantified in this demonstrator.

3.5 Possible Initiating Causes

The hazardous event can result from failures in several domains.

These causes are explored further in the FMEA and FTA rather than fully analysed here.

3.6 Hazard Analysis Summary

Field	Description
Operating situation	Humanoid walking slowly near a person while carrying a load. Knee is load-bearing.
Malfunctioning behaviour	Knee moves faster than commanded or produces unintended / excessive torque.
Hazardous event	Unintended knee motion occurs while load-bearing and close to a person, resulting in loss of controlled robot motion or posture.
Potential consequence	Collision, fall, dropped or destabilised load, injury to a person, or damage to equipment / robot.
Possible causes	Communication fault, controller fault, encoder fault, current-sensing fault, motor-control fault, inverter fault, transmission fault, safety-mechanism failure.
Required safety measures	Monitor commanded versus actual behaviour, cross-check sensing, supervise communication, limit torque / velocity, and provide a sufficiently independent fault-reaction path.
Safe reaction consideration	Immediate torque-off may itself create a hazard because the knee is load-bearing.

3.7 Safety Goal — SG-01

From HE-01, the following high-level safety goal is defined:

SG-01 — The knee joint system shall prevent or limit unintended or excessive joint torque during load-bearing operation such that hazardous uncontrolled knee motion is avoided.

3.8 Safe-State Consideration

A key outcome of the hazard analysis is that **torque-off cannot be assumed to be the universal safe state**.

For a load-bearing knee:

Immediate torque removal → loss of joint support → possible collapse

Depending on the detected fault and the remaining trustworthy functions, a safer reaction may instead be:

- torque limitation
- degraded operation

- controlled stop
- brake / hold
- independent torque disable
- robot-level recovery.

The final reaction therefore depends on which part of the control and actuation chain remains trustworthy after the fault.

3.9 Scope Limitation

This analysis does not assign a formal risk class or integrity level.

A production hazard analysis would require additional information such as:

- robot mass and payload,
- joint torque and velocity limits,
- human exposure conditions,
- operating modes,
- workspace assumptions,
- applicable machinery / robotics standards,
- severity and probability criteria.

For this demonstrator, the hazard analysis is used primarily to derive the safety goal and safety requirements and to support the subsequent FMEA, FTA and verification concept.

4. Safety Requirements

4.1 Purpose

The safety requirements are derived from the hazardous event **HE-01** and the associated safety goal **SG-01**.

The requirements below define the main detection, monitoring and fault-reaction functions needed to support this safety goal.

At this stage, exact numerical thresholds and reaction times are not assigned. These would need to be derived later from actuator dynamics, robot load cases and allowable fault-response time.

4.2 Requirement Derivation Approach

The requirements were derived by looking at the main ways HE-01 could develop through the architecture.

The intent is not only to detect individual component faults, but also to monitor whether the physical actuator behaviour remains consistent with the commanded behaviour.

4.3 Functional Safety Requirements

FSR-01 — Joint Velocity Monitoring

The knee joint system shall monitor the deviation between commanded joint velocity and measured joint velocity and detect when the deviation exceeds a defined allowable limit.

Rationale:

Unexpected joint velocity is one direct indication that the physical knee behaviour is no longer following the intended motion.

FSR-02 — Reaction to Excessive Velocity Deviation

When the commanded-to-measured joint velocity deviation exceeds the defined limit for a defined duration, the knee joint system shall initiate a safety reaction to limit or interrupt hazardous torque generation.

Rationale:

Detection alone is insufficient. A persistent motion deviation must result in a reaction before it develops into uncontrolled knee movement.

FSR-03 — Fault Reporting

The knee joint system shall report detected safety-relevant faults to the robot-level safety supervisor.

Rationale:

The local knee controller can detect actuator-level faults, but recovery may require whole-body actions

FSR-04 — Motor Current Monitoring

The knee joint system shall monitor the deviation between commanded motor current and measured motor current and detect when the deviation exceeds a defined allowable limit.

Rationale:

Motor current is directly related to torque generation. A significant commanded-versus-measured current deviation can indicate that the electrical actuation path is no longer behaving as requested.

FSR-05 — Reaction to Excessive Current Deviation

When a safety-relevant motor-current deviation is detected, the knee joint system shall initiate a defined fault reaction to prevent continued hazardous torque generation.

Rationale:

If actual current is no longer following the commanded current, normal torque control cannot automatically be assumed to remain trustworthy.

FSR-06 — Transmission Plausibility Monitoring

The knee joint system shall monitor the relationship between motor-side motion and joint-side motion and detect when the measured relationship deviates from the expected transmission ratio beyond a defined allowable limit.

Rationale:

Motor motion should have a predictable relationship with knee motion through the gearbox.

FSR-07 — Communication Supervision

The knee joint system shall detect missing, delayed, corrupted or stale safety-relevant commands received from the whole-body controller.

Rationale:

The local joint system cannot assume that every received command is fresh and valid.

FSR-08 — Reaction to Communication Fault

The knee joint system shall initiate a defined safe reaction when safety-relevant communication from the whole-body controller is missing, delayed, corrupted or stale.

Rationale:

Once the local joint can no longer rely on new robot-level commands, it should not continue unrestricted normal operation.

FSR-09 — Independent Safety Reaction Path

The knee joint system shall provide a safety reaction path that is sufficiently independent of the normal control path such that a single fault in the normal control path does not prevent mitigation of hazardous joint torque or motion.

Rationale:

If the normal joint controller itself generates the hazardous torque request, relying only on the same controller to remove that torque creates a single-point dependency.

FSR-10 — Sensor Fault Handling

The knee joint system shall, upon detection of an invalid or unavailable safety-relevant sensor signal, prevent unrestricted actuator operation and transition to a defined degraded mode or safe reaction based on the remaining valid sensing capability.

Rationale:

A single sensor fault does not necessarily require immediate torque removal. The reaction depends on whether enough trustworthy information remains to continue limited control safely.

4.4 Notes and Open Points

The following items remain to be derived during more detailed design:

- joint-velocity deviation threshold,
- current-deviation threshold,
- persistence times before declaring a fault,
- maximum diagnostic and reaction times,
- exact communication timeout,
- sensor accuracy and plausibility limits,
- required independence of the Safety Monitor,
- detailed degraded operating modes,
- exact torque-disable implementation.

These values should not be selected arbitrarily. They need to come from actuator dynamics, expected operating conditions, system-level response time and the consequences of delayed fault mitigation.

5. FMEA and Fault Tree Analysis

This section examines how faults in the knee-joint architecture can contribute to HE-01: hazardous unintended knee motion while the joint is load-bearing. The analysis is deliberately focused on the failure paths that matter to the demonstrator rather than attempting a complete production FMEA or FMEDA.

5.1 Analysis Scope

The analysis covers failures across sensing, communication, control, power electronics, mechanical transmission and the safety-reaction path. The aim is to understand how a local failure can propagate to hazardous joint behaviour and where detection or mitigation can interrupt that propagation.

A failure mode is retained in the portfolio analysis only when it contributes directly to HE-01, affects the ability to detect HE-01, or affects the ability to mitigate it. This keeps the analysis aligned with the project scope.

5.2 FMEA Method

The focused FMEA uses the following reasoning chain:

Failure Mode → Local Effect → System Effect → Detection → Fault Reaction

The distinction between local and system effect is important. The local effect describes what changes around the failed item. The system effect describes how that change can influence the knee joint or contribute to the hazardous event.

5.3 FMEA Observations

- Sensor faults can be hazardous even when the sensor output remains plausible.

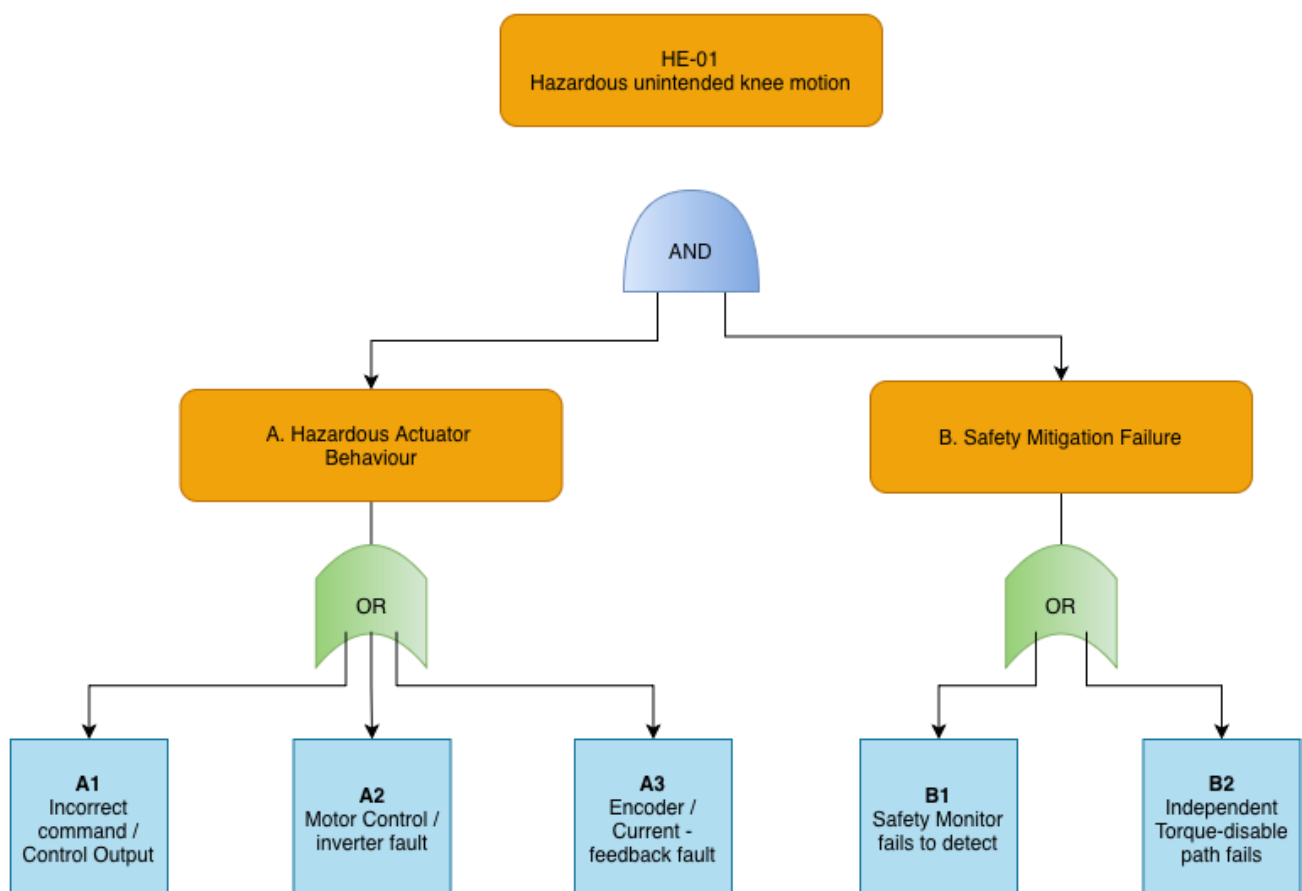
- Incorrect current feedback can result in either excessive torque or unintended torque reduction, depending on the failure direction.
- Motor-side and joint-side encoders provide different information and therefore support useful cross-checking across the gearbox.
- A local controller failure is more critical when the same controller is also required to execute the only available safety reaction.
- Power-stage faults can cause physical torque to differ from software intent, so monitoring only commands is insufficient.
- Mechanical transmission faults remain safety-relevant even when all electronics operate correctly.
- The safety monitor and independent torque-disable path must themselves be considered in the failure analysis.

5.4 Fault Tree Analysis Approach

The FTA complements the FMEA by starting from the hazardous system behaviour and reasoning backward toward possible causes. The tree is intentionally small and is used to show the main causal structure rather than quantify top-event probability.

5.5 Fault Tree Structure

Figure 5 - Simplified Fault Tree Analysis



The top event, HE-01, requires both:

Hazardous actuator behaviour AND failure of the available safety mitigation.

Hazardous actuator behaviour may result from incorrect command/control output, motor-control or inverter faults, or incorrect feedback.

Safety mitigation failure may result from failure of the Safety Monitor or the independent torque-disable path.

5.7 FTA Logic and Interpretation

The initiating faults on the left are connected by an OR gate because any one may cause hazardous actuator behaviour.

The mitigation failures on the right are also connected by an OR gate.

The top-level AND gate shows that HE-01 occurs when hazardous actuator behaviour is present and the intended mitigation does not successfully control it.

5.8 Link to Safety Requirements

The FMEA and FTA support the requirements derived in Section 4. The main relationships are:

- FSR-01 / FSR-02 address hazardous joint-velocity deviation and the required reaction.
- FSR-04 / FSR-05 address unintended motor-current behaviour and torque-generation faults.
- FSR-06 addresses inconsistency between motor-side and joint-side motion.
- FSR-07 / FSR-08 address invalid or unavailable robot-level commands.
- FSR-09 addresses the possibility that the normal control path itself is faulty.
- FSR-10 addresses invalid or unavailable safety-relevant sensing.

5.9 Limitations

This is a qualitative system-level analysis. It does not include component failure rates, FIT data, diagnostic coverage calculations, quantitative probability evaluation, or a complete dependent-failure analysis. These would require a more detailed hardware design and reliability data.

The purpose of this analysis is to demonstrate the link between architecture, failure propagation, safety mechanisms and requirements for the selected hazardous event.

5.10 Related Project Artifacts

- 05_FMEA.xlsx — full focused FMEA working table
- 05_fault_tree.drawio — editable fault-tree source
- 05_fault_tree.png or .svg — exported fault-tree figure for GitHub and the final report

6. Safety Concept, Traceability and Verification

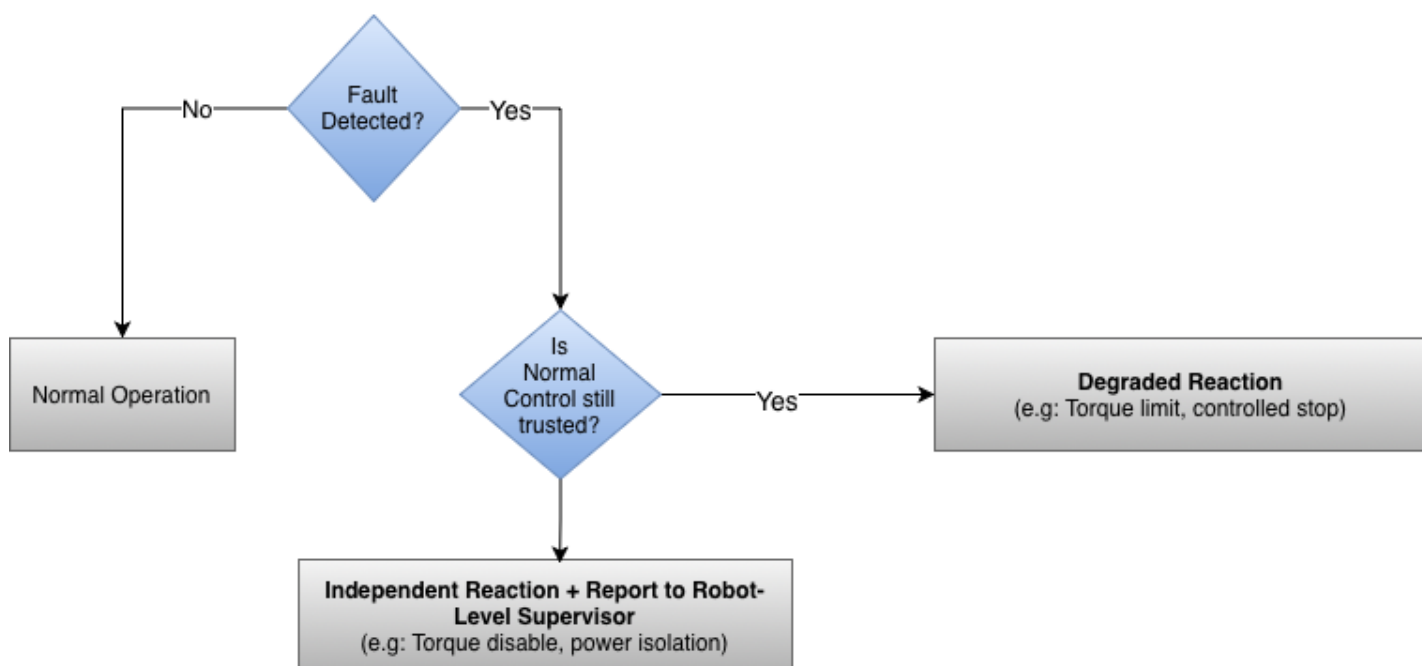
6.1 Purpose

This section defines the fault-reaction concept and links the safety requirements to architecture and verification.

Safety focus: HE-01 - hazardous unintended knee motion while the joint is load-bearing. All concepts below trace back to SG-01 and the ten FSRs defined earlier.

6.2 Safety Concept and Fault-Reaction Logic

Figure 6 - Fault-Reaction Decision Concept



The safety concept is based on the amount of control capability that remains trustworthy after a fault. A communication fault, sensor fault and inverter fault therefore do not automatically result in the same reaction.

If local control and sensing remain trustworthy, the preferred response is to restrict capability and move toward a controlled stop. If the normal control or torque-generation path can no longer be trusted, the reaction must use an independent mechanism.

Immediate torque-off is not treated as the default response because loss of knee support may itself destabilise the robot.

6.3 Fault-Reaction Matrix

Detected condition	Trusted capability	Reaction
Communication unavailable / invalid	Local control and sensing healthy	Controlled stop, then local hold / degraded state
Safety-relevant sensor fault	Remaining sensing and controller healthy	Restrict torque/velocity; degraded mode or controlled stop
Joint motion / transmission inconsistency	Control responsive, mechanical state uncertain	Torque limitation + controlled stop / robot-level recovery
Local joint controller untrusted	Independent safety path healthy	Independent torque disable
Gate driver / inverter untrusted	Safety monitor and independent isolation healthy	Gate disable / power isolation + coordinated recovery
Safety path itself unavailable	Normal control healthy but protection degraded	Prevent unrestricted operation and perform controlled robot-level stop

6.4 Independent Safety Path

The normal controller may itself be the source of hazardous torque. The architecture therefore includes a hardware-capable reaction path that can inhibit torque generation without relying solely on the normal control software.

The final implementation must address common dependencies such as shared power, processor, clock and gate-control resources.

6.5 Requirements-to-Architecture Traceability

The table below shows how each functional safety requirement is implemented conceptually in the architecture and how it will be verified. HE-01 / SG-01 apply to all rows.

FSR	Architecture Element	Safety Mechanism	V&V Ref
FSR-01	Joint Encoder + Safety Monitor	Velocity-deviation monitoring	VVT-01
FSR-02	Safety Monitor + Motor Control	Torque/velocity limitation	VVT-01
FSR-03	Safety Monitor + Communication Interface	Fault-status reporting	VVT-07
FSR-04	Current Sensors + Safety Monitor	Current-deviation monitoring	VVT-02
FSR-05	Safety Monitor + Gate/Disable Path	Current limitation / torque inhibit	VVT-02
FSR-06	Motor Encoder + Joint Encoder	Transmission plausibility	VVT-03

FSR	Architecture Element	Safety Mechanism	V&V Ref
FSR-07	Communication Interface	Timeout / integrity supervision	VVT-04
FSR-08	Local Controller + Safety Monitor	Invalid-command rejection / controlled reaction	VVT-04
FSR-09	Independent Torque-Disable Path	Independent torque inhibit	VVT-06
FSR-10	Sensors + Safety Monitor	Sensor plausibility / degraded reaction	VVT-05

6.6 Verification Strategy

Verification is planned progressively. SiL/HiL testing is used for diagnostic logic and fault injection; actuator bench testing covers the real motor, inverter and sensors; integration testing covers interaction with the robot-level safety supervisor. Analysis is used for timing, independence and common-cause assumptions

6.7 V&V Matrix

V&V ID	Verification objective	Requirements	Method	Pass concept
VVT-01	Joint velocity deviation monitoring and reaction	FSR-01, FSR-02	SIL / HIL fault injection	Inject excessive commanded-vs-measured velocity deviation; detect and initiate defined reaction within required time.
VVT-02	Motor current monitoring and reaction	FSR-04, FSR-05	HIL / bench test	Inject incorrect current feedback or emulate overcurrent; detect and limit / disable as defined.
VVT-03	Motor-to-joint transmission plausibility	FSR-06	SIL / HIL	Inject encoder offset or simulated gearbox slip; detect mismatch above allowable tolerance.
VVT-04	Communication supervision and reaction	FSR-07, FSR-08	Communication fault injection	Drop, delay, repeat and corrupt commands; reject invalid data and enter defined reaction.
VVT-05	Sensor fault handling	FSR-10	SIL / HIL fault injection	Freeze, offset or remove safety-relevant sensor signals; prevent unrestricted actuator operation.

V&V ID	Verification objective	Requirements	Method	Pass concept
VVT-06	Independent safety reaction path	FSR-09	HIL / integration	Force a main-controller fault while hazardous torque is requested; verify independent path still inhibits torque.
VVT-07	Fault reporting to robot safety supervisor	FSR-03	Integration test	Inject representative safety faults; verify correct fault/status message arrives within required reporting time.

6.8 Open Points and Limitations

- Diagnostic thresholds, persistence times and reaction-time budgets still need quantitative derivation.
- Safety Monitor independence and the torque-disable implementation require detailed hardware/software design.
- Quantitative reliability, diagnostic coverage and common-cause analysis are outside the current demonstrator scope.
- Verification activities are planned; physical test evidence has not yet been generated.